

Air University



Operating Systems (Lab)

Year 2025

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Academic Year: 2024 – 2028

Department of Cyber Security

USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

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Lab Task # 01

Find and execute the following commands:

- show the architecture of the machine

Command: systeminfo | findstr /I "System Type"

```
PS D:\> systeminfo | findstr /I "System Type"
OS Build Type:           Multiprocessor Free
System Boot Time:        03-Feb-26, 1:52:45 PM
System Manufacturer:     HP
System Model:             HP EliteBook 830 G6
System Type:             x64-based PC
System Directory:         C:\WINDOWS\system32
System Locale:            en-us;English (United States)
```

- show CPU info

Command: Get-CimInstance Win32_Processor | Select-Object Name

```
PS D:\> Get-CimInstance Win32_Processor | Select-Object Name
Name
----
Intel(R) Core(TM) i7-8665U CPU @ 1.90GHz
```

- show system date

Command: Get-Date -Format "dd/MM/yyyy HH:mm:ss"

```
PS D:\> Get-Date -Format "dd/MM/yyyy HH:mm:ss"
06-02-2026 11:29:50
```

- show details of files and directories

Command: dir filename

```
PS D:\> dir University

Directory: D:\University

Mode                LastWriteTime         Length Name
----                -
d-----          01-Feb-26   2:37 PM           Cryptography
d-----          02-Feb-26   9:28 PM       Multivariable Calculus
d-----          01-Feb-26   2:37 PM       Network Security
d-----          03-Feb-26   8:27 PM       Operating Systems
d-----          03-Feb-26   8:33 PM Secure Software Design & Development
```

Get-

Command: ChildItem "d:\University"

```
PS D:\> Get-ChildItem "d:\University"

Directory: D:\University

Mode                LastWriteTime         Length Name
----                -
d-----          01-Feb-26   2:37 PM           Cryptography
d-----          02-Feb-26   9:28 PM       Multivariable Calculus
d-----          01-Feb-26   2:37 PM       Network Security
d-----          03-Feb-26   8:27 PM       Operating Systems
d-----          03-Feb-26   8:33 PM Secure Software Design & Development
```

- show hidden files

Command: Get-ChildItem -hidden

```
PS D:\> Get-ChildItem -hidden

Directory: D:\

Mode                LastWriteTime         Length Name
----                -
d--hs-            12-Oct-25   4:15 PM             $RECYCLE.BIN
d--hs-            07-Oct-24  12:22 AM      System Volume Information
```

- create a directory called 'dir1'

Command: mkdir dir1

```
PS D:\University\Operating Systems\Lab\Lab 1> mkdir dir1

Directory: D:\University\Operating Systems\Lab\Lab 1

Mode                LastWriteTime         Length Name
----                -
d-----            06-Feb-26  11:39 AM             dir1
```

- show files and directories containing numbers

Command: dir | findstr [0-9]

```
PS D:\University\Operating Systems\Lab\Lab 1> dir | findstr [0-9]

Directory: D:\University\Operating Systems\Lab\Lab 1

d-----            06-Feb-26  11:39 AM             dir1
d-----            06-Feb-26  10:06 AM             new
-a-----            06-Feb-26   9:47 AM        64699 cmd commands.pdf
-a-----            06-Feb-26  11:20 AM        196679 commands.pdf
-a-----            06-Feb-26  10:17 AM       3632547 Lab Task 1 - 247141.docx
-a-----            06-Feb-26  10:18 AM       335100 Lab Task 1 - 247141.pdf
```

- create two directories simultaneously

Command: New-Item -itemType Directory dir2,dir3

```
PS D:\University\Operating Systems\Lab\Lab 1> New-Item -itemType Directory dir2,dir3

Directory: D:\University\Operating Systems\Lab\Lab 1

Mode                LastWriteTime         Length Name
----                -
d-----            06-Feb-26  11:42 AM             dir2
d-----            06-Feb-26  11:42 AM             dir3
```

- show the path of the work directory

Command: pwd

```
PS D:\University\Operating Systems\Lab\Lab 1> pwd

Path
----
D:\University\Operating Systems\Lab\Lab 1
```

- delete file called 'file1'

Command: del 'filename'

```
PS D:\University\Operating Systems\Lab\Lab 1> del 'Lab Task 1 - 247141.pdf'
PS D:\University\Operating Systems\Lab\Lab 1> dir
```

Directory: D:\University\Operating Systems\Lab\Lab 1

Mode		LastWriteTime	Length	Name
d-----		06-Feb-26 11:39 AM		dir1
d-----		06-Feb-26 11:42 AM		dir2
d-----		06-Feb-26 11:42 AM		dir3
d-----		06-Feb-26 10:06 AM		new
-a----		06-Feb-26 9:47 AM	64699	cmd commands.pdf
-a----		06-Feb-26 11:20 AM	196679	commands.pdf
-a----		06-Feb-26 10:17 AM	3632547	Lab Task 1 - 247141.docx

- remove a directory called 'dir1'

Command: Remove-Item dir2 -Recurse -Force

```
PS D:\University\Operating Systems\Lab\Lab 1> Remove-Item dir2 -Recurse -Force
PS D:\University\Operating Systems\Lab\Lab 1> dir
```

Directory: D:\University\Operating Systems\Lab\Lab 1

Mode		LastWriteTime	Length	Name
d-----		06-Feb-26 11:42 AM		dir3
d-----		06-Feb-26 10:06 AM		new
-a----		06-Feb-26 9:47 AM	64699	cmd commands.pdf
-a----		06-Feb-26 11:20 AM	196679	commands.pdf
-a----		06-Feb-26 10:17 AM	3632547	Lab Task 1 - 247141.docx

- delete directory called 'dir1' that is empty

Command: rmdir dir1

```
PS D:\University\Operating Systems\Lab\Lab 1> rmdir dir1
PS D:\University\Operating Systems\Lab\Lab 1> dir
```

Directory: D:\University\Operating Systems\Lab\Lab 1

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	06-Feb-26 11:42 AM		dir2
d-----	06-Feb-26 11:42 AM		dir3
d-----	06-Feb-26 10:06 AM		new
-a----	06-Feb-26 9:47 AM	64699	cmd commands.pdf
-a----	06-Feb-26 11:20 AM	196679	commands.pdf
-a----	06-Feb-26 10:17 AM	3632547	Lab Task 1 - 247141.docx

- modify the timestamp of a file or directory

Command: (Get-Item dir3).LastWriteTime='2020-02-20 01:30'

```
PS D:\University\Operating Systems\Lab\Lab 1> (Get-Item dir3).LastWriteTime='2020-02-20 01:30'
PS D:\University\Operating Systems\Lab\Lab 1> dir
```

Directory: D:\University\Operating Systems\Lab\Lab 1

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	20-Feb-20 1:30 AM		dir3
d-----	06-Feb-26 10:06 AM		new
-a----	06-Feb-26 9:47 AM	64699	cmd commands.pdf
-a----	06-Feb-26 11:20 AM	196679	commands.pdf
-a----	06-Feb-26 10:17 AM	3632547	Lab Task 1 - 247141.docx

THE END
